

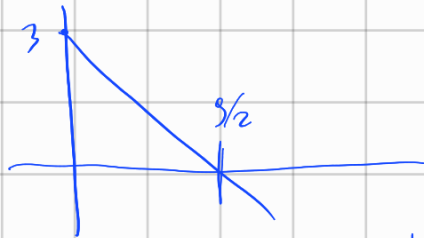
1° APPELLO 2022

$$① \iint_{\Sigma} (x+z) d\sigma \quad \Sigma = \{2x+3y+6z=9\} \quad z = \frac{3}{2} - \frac{1}{2}y - \frac{1}{3}x$$

$$d\sigma = \sqrt{1 + \left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{3}\right)^2} = \frac{7}{6}$$

$$\frac{7}{6} \iint_{\Sigma} x + \frac{3}{2} - \frac{1}{2}y - \frac{1}{3}x d\alpha d\gamma = \frac{7}{6} \iint_{\Sigma} \frac{2}{3}x - \frac{1}{2}y + \frac{3}{2} d\alpha d\gamma =$$

$$z=0 \rightarrow 2x+3y=9 \rightarrow y = 3 - \frac{2}{3}x$$



$$\int_0^{\frac{9}{2}} \int_0^{3-\frac{2}{3}x} \left(\frac{2}{3}x - \frac{1}{2}y + \frac{3}{2} \right) d\alpha d\gamma = \int_0^{\frac{9}{2}} \left| \frac{2}{3}x\gamma - \frac{1}{4}\gamma^2 + \frac{3}{2}\gamma \right|_0^{3-\frac{2}{3}x} d\alpha =$$

$$= \int_0^{\frac{9}{2}} \left(\frac{2}{3}x \left(3 - \frac{2}{3}x \right) - \frac{1}{4} \left(3 - \frac{2}{3}x \right)^2 + \frac{3}{2} \left(3 - \frac{2}{3}x \right) \right) d\alpha =$$

$$= \frac{7}{6} \int_0^{\frac{9}{2}} \left(2x - \frac{4}{9}x^2 - \frac{9}{4} + \frac{1}{9}x^2 + \frac{9}{2} - x \right) d\alpha = \frac{7}{6} \int_0^{\frac{9}{2}} \left(x - \frac{3}{9}x^2 + \frac{9}{4} \right) d\alpha =$$

$$= \frac{7}{6} \left| \frac{x^2}{2} - \frac{1}{9}x^3 + \frac{9}{4}x \right|_0^{\frac{9}{2}} = \frac{7}{6} \cdot \left(\frac{81}{4} - \frac{135}{8} + \frac{81}{8} \right) = \frac{7}{6} \cdot \frac{108}{8} = \frac{63}{4}$$

6
2

$$\iint_A \left(\frac{1}{xy} \right) dx dy \quad A = \{0 < x < y < 3x, 1 < x+y < 2\}$$

$$\begin{cases} \frac{y}{x} = u \\ x+y = v \end{cases} \begin{cases} y = u(v-y) \\ x = v-y \end{cases} \begin{cases} y = uv - uy \\ / \end{cases} \begin{cases} y = \frac{uv}{1+u} \\ x = \frac{v}{1+u} \end{cases}$$

$$J = \begin{pmatrix} \frac{-v}{(1+u)^2} & \frac{1}{(1+u)} \\ \frac{v}{(1+u)^2} & \frac{u}{(1+u)} \end{pmatrix} = \frac{-v \cdot u - v}{(1+u)^3} = \frac{v}{(1+u)^2}$$

$$\iint_A \frac{1}{xy} dx du = \iint_E \frac{(1+u)^2}{u v^2} \cdot \frac{v}{(1+u)^2} du dv \quad E = \{(u,v) \in \mathbb{R}^2 \mid 1 \leq u \leq 3, 1 \leq v \leq 2\}$$

$$\int_1^2 \int_1^3 \frac{1}{u \cdot v} du dv = \log(3) \cdot \log(2)$$

$\underbrace{\quad}_{\frac{1}{4}}$

$$rot = x^2 + 2 - x^2 = 2$$

$$2 \int_0^{\frac{\pi}{2}} \int_0^{\cos \theta} p \, dp \, d\theta = 2 \int_0^{\frac{\pi}{2}} \frac{(\cos \theta)^2}{2} = \frac{\pi}{4} + 0 = \frac{\pi}{4}$$

$$5) f(x,y) = x^2y - 2x^2 - y^2 \quad \nabla f = (2xy - 4x, x^2 - 2y)$$

$$f_{xx} = 2y - 4 \quad f_{yy} = -2 \quad f_{xy} = 2x \quad f_{yx} = 2x$$

$$\begin{cases} 2x(y-2) = 0 \\ x^2 = 2y \end{cases} \begin{cases} A(0,0) \\ B(2,2) \\ C(-2,2) \end{cases} \quad \nabla^2 f(x,y) = \begin{pmatrix} 2(y-2) & 2x \\ 2x & -2 \end{pmatrix}$$

$B \in C$ $\det < 0$ saddle $A \rightarrow \det > 0, f_{xx} < 0$: max

↳ 3

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1 \quad y = (1 - x^{\frac{2}{3}})^{\frac{3}{2}} \quad y' = -\frac{2}{3} \cdot x^{-\frac{1}{3}} \cdot (1 - x^{\frac{2}{3}})^{\frac{1}{2}} \cdot \frac{3}{2}$$

$$(\dot{y})^2 = x^{-\frac{2}{3}} (1 - x^{\frac{2}{3}}) = x^{-\frac{2}{3}} - 1 \quad \sqrt{1 + f'^2} = x^{-\frac{1}{3}}$$

$$2 \int_0^1 x^{-\frac{1}{3}} = 3$$